

5.2 Application Development Platform

Many software, application and games are developed for mobiles phones and PDAs using application development platforms such as Java and BREW. Applications on a mobile phone are generally developed by the manufacturer of the device, or a third party. You, too, can develop applications and games for your mobile. All you need is a PC-to-mobile interface and in-depth knowledge of programming and your device's architecture. But, if the software you develop doesn't match the device architecture, it will hamper the functioning of the device.

5.2.1 JAVA Platform—Micro Edition

Java ME or J2ME is a set of JAVA APIs meant for programming of an embedded system device such as mobile phones and PDAs. Java has become increasingly popular for creating games and applications for such devices because it is platform-independent, and has specific set of profiles and configuration that adhere to a family of device. For example, 'Mobile Information Device Profile' is targeted at mobile devices such as cell phones, and 'Personal Profile' is meant for consumer devices such as set-top boxes and PDAs.

Profiles form a superset of Configurations. A Configuration is a set of Java classes and libraries. There are two types of Configurations: Connected Limited Device Configuration and Connected Device Configuration. The former contains a strict set of libraries and Java classes, which is minimal requirement for a Java Virtual Machine to operate; the latter contains almost all the libraries that are not GUI-related.

J2ME also includes a modular set of Optional Packages that offer standard APIs for using both existing and emerging technologies such as database connectivity, multimedia, Bluetooth and Web services. The modular structure of these packages helps avoid unnecessary functionality by including only those packages and applications actually required by a device.